Study Questions: Communication

1. What are mirror neurons? How did Rizzolatti et al. (1996) discover them?
2. How does Active Inference interpret mirror neuron activity as motor simulation and prediction?
3. What is interpersonal neural synchrony, and how is it measured (hyperscanning)?
4. Describe Hasson et al.'s (2012) speaker-listener coupling study. What correlated with successful communication?
5. How does Active Inference's concept of "generalized synchrony" map onto interpersonal neural coupling?
6. What brain regions comprise the mentalizing network (Theory of Mind network)? What does each contribute?
7. How does the temporo-parietal junction (TPJ) contribute to reasoning about others' beliefs?
8. Compare the mirror neuron system (motor simulation) with the mentalizing network (belief attribution). Are they complementary or competing accounts?
9. How does Active Inference explain the development of Theory of Mind in children (typically around age 4)?
10. What is the "shared manifold hypothesis" (Gallese) and how does it relate to Active Inference's account of communication?
11. How might reduced mirror neuron activity contribute to social communication difficulties in autism?
12. What critical evidence challenges the mirror neuron theory of social cognition?
13. How does Active Inference explain the difference between sarcasm detection (requiring ToM) and literal comprehension?
14. What role does prosody (tone of voice) play in social inference from an Active Inference perspective?
15. How does the concept of "joint attention" (Tomasello) relate to coupled inference between caregiver and infant?
16. Explain how paranoid ideation in schizophrenia might result from overactive mentalizing (excessive ToM).

17. How does language comprehension involve prediction at multiple levels (phoneme, word, sentence, discourse)?
18. What does the neural synchrony literature suggest about the role of rhythm and timing in communication?
19. How does social media communication differ from face-to-face interaction in terms of the available channels for neural coupling?
20. Design a hyperscanning study that could test whether increased speaker-listener neural coupling predicts persuasion effectiveness.